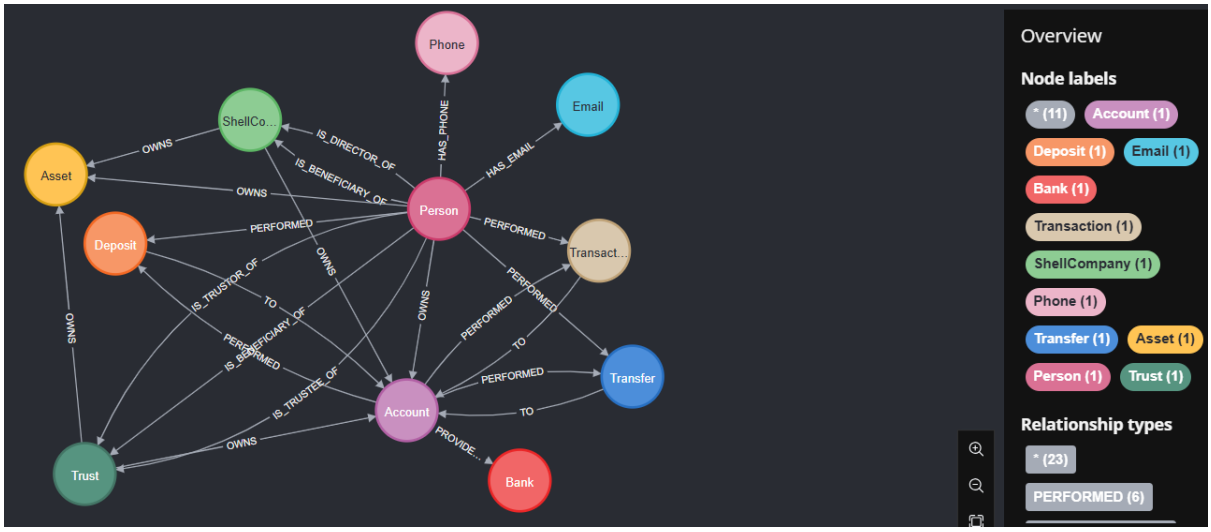
Lab 4

Exercise 1.1 Neo4j money Laundering dataset analysis

```
CALL
apoc.cypher.runFile("https://gist.githubusercontent.com/apogiatzis/7e53f828f0566980b935b314d575226d/raw/682a030f9c24a46e67e6481ea66c484bca54c5a4/lab3_ex_import.cypher")

CALL db.schema.visualization();
```




```
MATCH (n)
RETURN head(labels(n)) as type, count(*) as count
```

	type	count
1	"Email"	156
2	"Phone"	155
3	"Person"	159
4	"Trust"	16
5	"ShellCompany"	27
6	"Bank"	10
7		

ted streaming 9 records after 104 ms and completed after 197 ms.

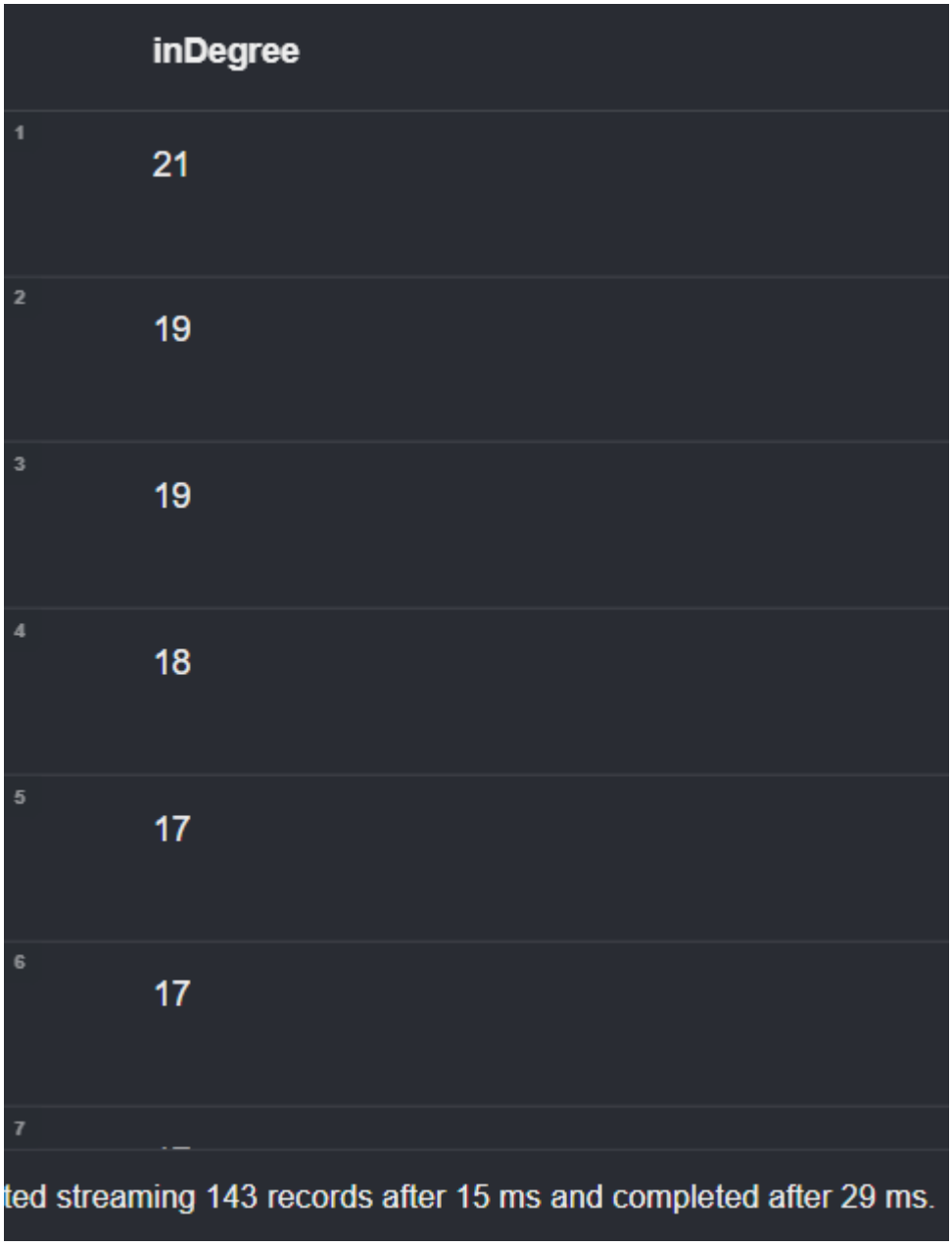
```
MATCH (p:Account)
WITH apoc.node.degree(p,"<") AS inDegree, apoc.node.degree(p,">") AS
outDegree
```

```
RETURN inDegree, outDegree
ORDER BY inDegree DESC;
```

	inDegree	outDegree
1	22	2
2	20	4
3	20	18
4	19	10
5	18	14
6	18	14
7		

rted streaming 143 records after 18 ms and completed after 36 ms.

```
MATCH (p:Account)
RETURN apoc.node.degree(p,"<TO") AS inDegree
ORDER BY inDegree DESC;
```



	inDegree
1	21
2	19
3	19
4	18
5	17
6	17
7	

ted streaming 143 records after 15 ms and completed after 29 ms.

```
MATCH (p:Account)
RETURN apoc.node.degree(p,"<TO") AS inDegree, p.obj_id
ORDER BY inDegree DESC;
```

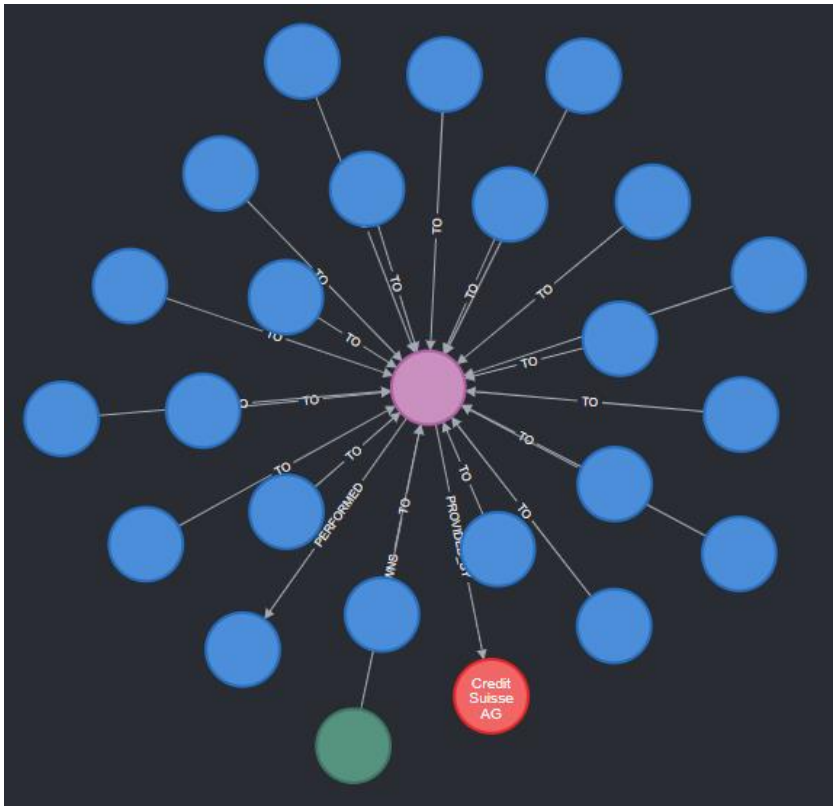
	inDegree	p.obj_id
1	21	"38f2a17e-88a9-45ee-acab-8dbfd125241e"
2	19	"eb2b40b6-2276-4071-8700-44b153b3d02c"
3	19	"b3a00e8a-457b-4afc-b860-d11cf8309628"
4	18	"171a434e-97fd-4eeb-a6f0-aa950eb87112"
5	17	"da6df937-c734-4b81-bb55-cb074c27368d"
6	17	"73205398-474a-40f0-af70-7d2498e5f706"
7		

rted streaming 143 records after 16 ms and completed after 88 ms.

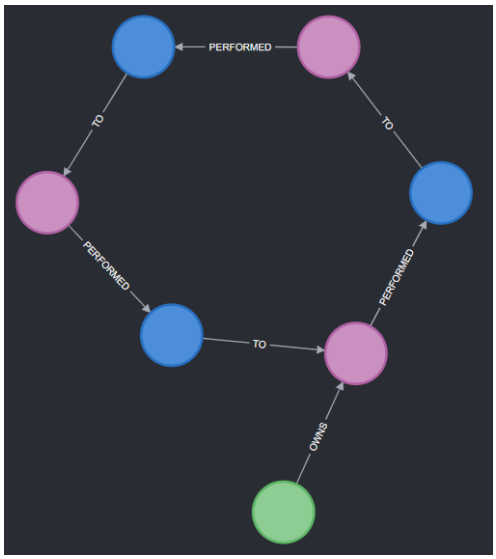
```
MATCH (p:Account)<-[:TO]-(t:Transfer)
WHERE p.obj_id = "38f2a17e-88a9-45ee-acab-8dbfd125241e"
WITH min(t.timestamp) as minDate, max(t.timestamp) as maxDate
WITH *, (maxDate.epochMillis-minDate.epochMillis)/1000/60/60 as
hoursTimeDelta
RETURN minDate, maxDate, hoursTimeDelta
```

	minDate	maxDate	hoursTimeDelta
1	"2021-03-02T11:19:00Z"	"2021-03-03T07:17:00Z"	19

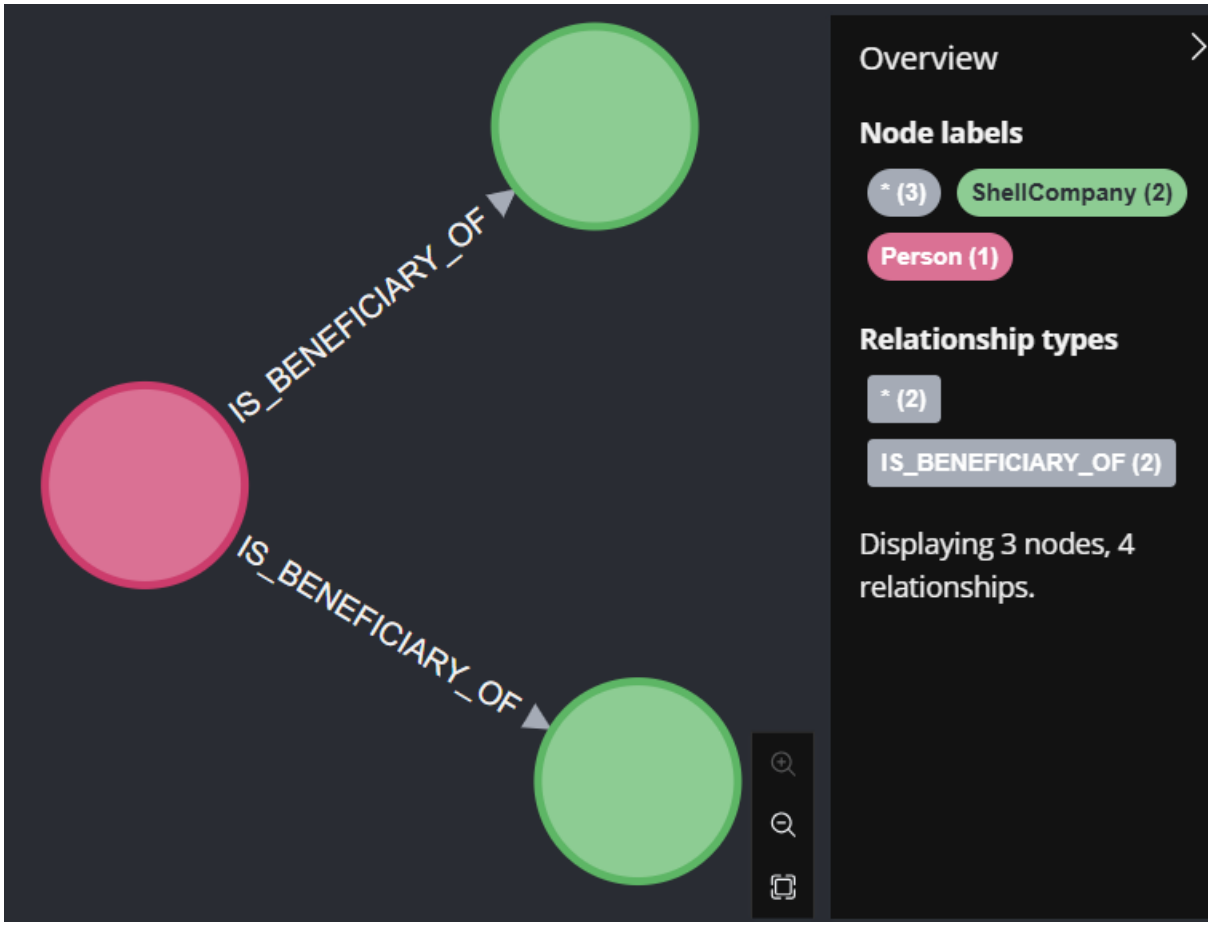
```
MATCH (p:Account)
RETURN apoc.node.degree(p,"<TO") AS inDegree, p
ORDER BY inDegree DESC
LIMIT 1;
```



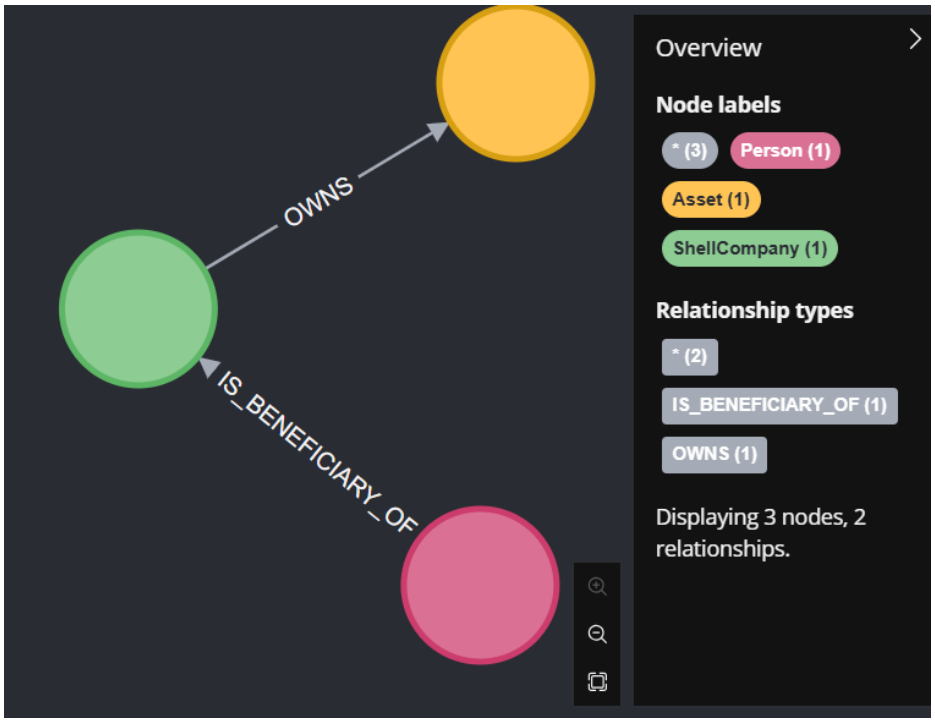

```
MATCH p=(s:ShellCompany {obj_id:"9e5ae7d7-2482-40be-9e05-4cd55c782060"})-[:OWNS]-(a:Account)-[*]->(a)
RETURN nodes(p)
```




```
WITH ["4cf2176b-d5ee-4409-a30c-3c9ca1639f61", "ebb647bf-45f2-4246-868d-4ed9d2a30449", "
9e5ae7d7-2482-40be-9e05-4cd55c782060"] as shellCompaniesIds
MATCH p=(s1:ShellCompany)-[*..1]-()-[*..1]-(s2:ShellCompany)
WHERE s1.obj_id IN shellCompaniesIds AND s2.obj_id IN shellCompaniesIds AND s1.obj_id <> s2.obj_id
RETURN p
```




```
MATCH path=(p:Person)-[*]->()-[:OWNS]->(a:Asset)
WHERE p.obj_id = "580fc4ad-8b88-4f4b-9ef1-19ea974eafa3" AND
toFloat(a.value_in_gbp) > 10000
RETURN path
```



```
path
(:Person {obj_id: "580fc4ad-8b88-4f4b-9ef1-19ea974eafa3",occupation: "
Armed forces operational officer",nationality: "Isle of Man",phone: "+
44(0)28 9018 0046",date_of_birth: "1983-02-27",ni_number: "ZZ 60 75 99
 T",last_name: "Brown",first_name: "Malcolm"})-[:IS_BENEFICIARY_OF]->(
:ShellCompany {obj_id: "9e5ae7d7-2482-40be-9e05-4cd55c782060"})-[:OWNS
]->(:Asset {obj_id: "b9c77a05-5819-4d1a-9cc5-9d761c16760a",value_in_gb
p: "2500000.0",asset_type: "PHYSICAL",description: "Silver superyacht"
})
```